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RESEARCH PRESENTATION

A Speed-Gradient Operator for Asymptotic Static State Recovery in Unsteady Viscous Flows

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Problem: static loading at one mean angle from pitching data

Measured

Surface pressure over full pitch cycles about one mean angle α_0.

Wanted

Static loading at α_0 only, as if the airfoil were held fixed there.

Obstacle and contribution

Phase averaging leaves a nonlinear mean bias. Harmonic fitting plus extrapolation across amplitudes ε removes it.

$$\alpha(t) = \alpha_0 + \varepsilon \sin(\omega t) \quad \Rightarrow \quad \Delta C_{p0}(x; \alpha_0)$$

pp. 1–2, Introduction; p. 2, Sec. 2.1 and Eq. (1)

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Why a plain average is off: a nonlinear mean bias remains

$$\Delta C_p = \Delta C_{p0} + \varepsilon \Delta C_{p1} + \varepsilon^2 \Delta C_{p2} + \mathcal{O}(\varepsilon^3)$$

Piece	How it varies in time	Cycle average
Static loading	Constant	ΔC_{p0}
Main swing	$\sin(\omega t),\ \cos(\omega t)$	0
Nonlinear part	$D_0 + \text{second harmonic}$	$\varepsilon^2 D_0$

Average over a cycle = static loading + a small offset (mean bias)

p. 2, Eqs. (2)–(4)

Step 1: the speed-gradient operator isolates the constant part

$$y(t) \approx c_0 + c_1 \cos(\omega t) + s_1 \sin(\omega t) + \cdots$$



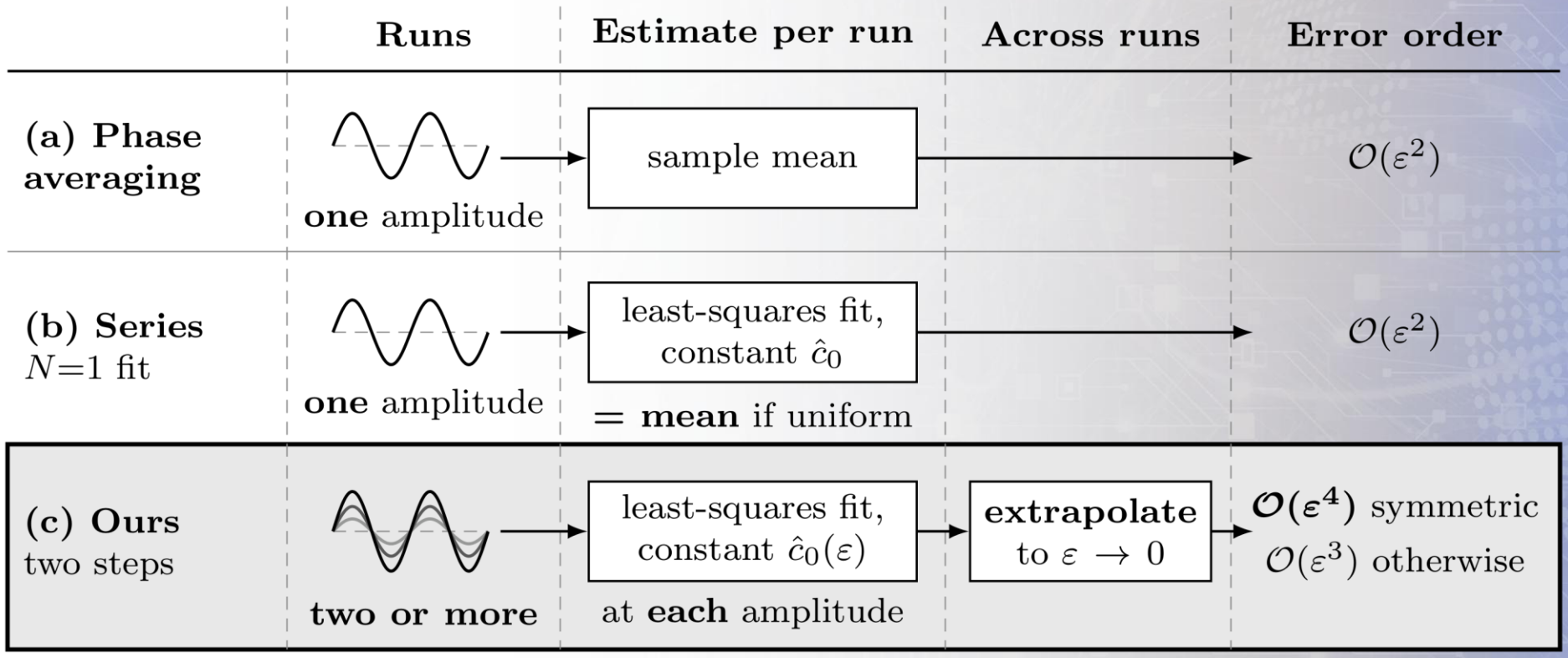
$$\hat{c}_0 = \mathbf{e}_1^{\mathsf{T}} (\mathbf{\Phi}^{\mathsf{T}} \mathbf{\Phi})^{-1} \mathbf{\Phi}^{\mathsf{T}} \mathbf{y}$$

This formula is the speed-gradient operator: a least-squares wave fit that returns the constant c_0.

"Speed": set by the reduced frequency k (pitch speed vs flow speed). "Gradient": separates steady from oscillating.

p. 2, Theorem 1 and Eqs. (5)–(6); p. 3, Eq. (7)

With even full-cycle samples, the fit equals a plain average



The reported accuracy gain comes from combining amplitudes

p. 3, Secs. 3.3.1 and 3.4; p. 4, Sec. 5.1

Step 2: combining two amplitudes cancels the mean bias

$$\hat{c}_0(\varepsilon) = \Delta C_{p0} + \varepsilon^2 D_0 + \text{higher-order terms}$$

$$\widehat{\Delta C}_{p0}^{(2)}(x_j) = \frac{\varepsilon_2^2 \hat{c}_0(\varepsilon_1) - \varepsilon_1^2 \hat{c}_0(\varepsilon_2)}{\varepsilon_2^2 - \varepsilon_1^2}$$

Two amplitudes: the bias ($\propto \varepsilon^2$) cancels, the static loading remains

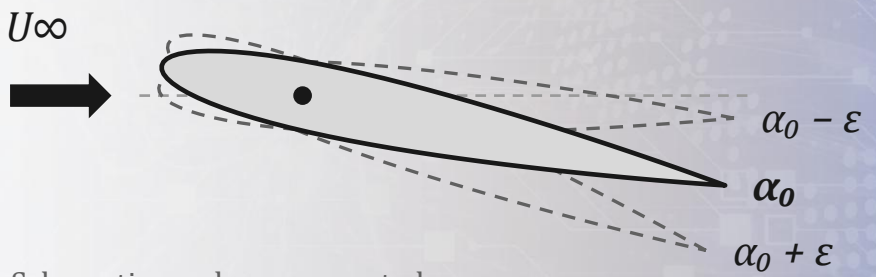
$$\mathcal{O}(\varepsilon^4) \text{ in the symmetric case;} \quad \mathcal{O}(\varepsilon^3) \text{ otherwise}$$

Top: fitted constant = static loading + bias. Middle: the combination of two runs that cancels it.

p. 3, Sec. 3.3.2 and Eq. (8); p. 3, Table 1

How to pitch the airfoil: one mean angle, several amplitudes

1. Set the mean angle α_0 to the angle whose static loading is wanted
2. Pitch harmonically about it: $\alpha(t) = \alpha_0 + \varepsilon \sin(\omega t)$
3. Sample pressure at evenly spaced phases over full cycles (≥ 5 per cycle)
4. Repeat at two or more amplitudes ε (paper: five, from 0.5° to 5°)



Schematic; angles exaggerated



Same α_0, three amplitudes ε; circles mark evenly spaced samples

Fit the constant at each amplitude, then extrapolate to ε → 0

p. 2, Sec. 2.1 and Eq. (1); p. 3, Table 1; p. 5, Sec. 5.2 and Table 3

Two computer-generated tests: algebra check, viscous model

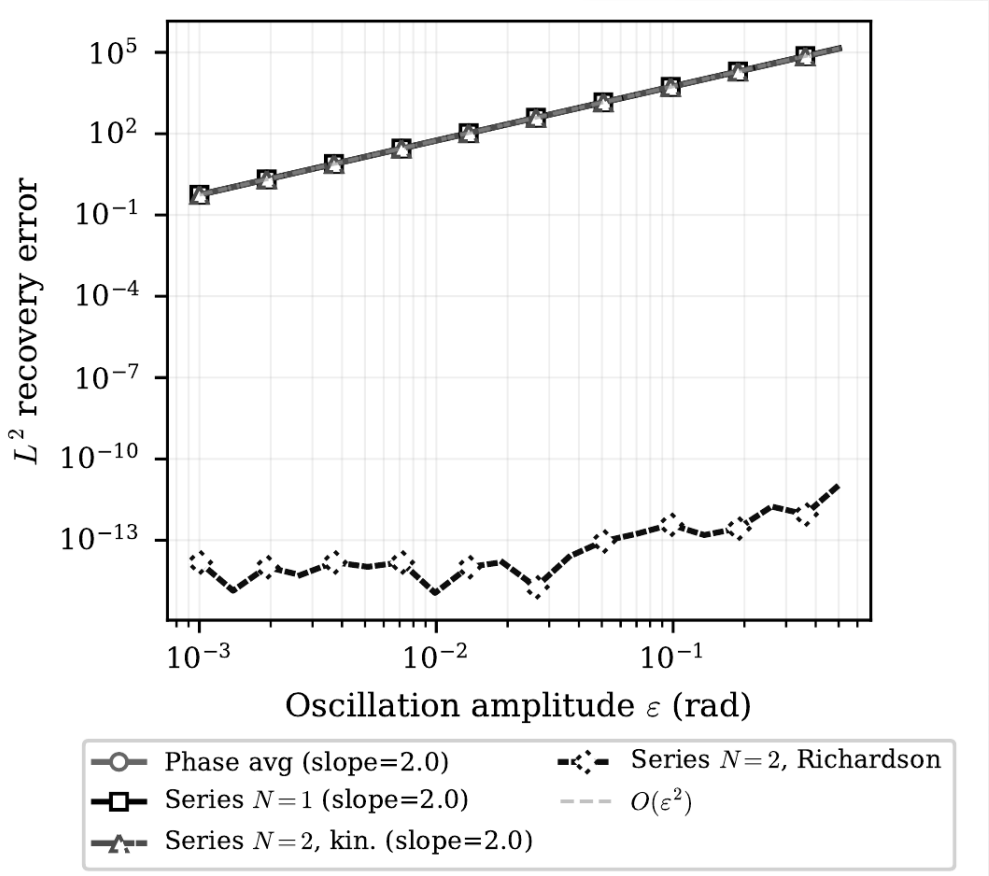
	Test 1: synthetic, ideal flow	Test 2: viscous, slow motion
Data source	Second-order Theodorsen signal	NeuralFoil (trained on XFOIL)
Reduced frequency	$k = 0.15$	$k \to 0$
Sampling	64 phases; 500 chord stations	32 phases; 32 chord stations
What it tests	Is the algebra right?	Is the bias removed?

Both data sets are computer-generated. Fast motion with viscosity is the next test.

p. 3, Table 2 and Sec. 4.1; p. 4, Secs. 4.2–4.3

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Synthetic test: the correction removes the bias completely



Test signal matches the method's assumptions

Plain average, basic fit: error ∝ amplitude² (slope 2)

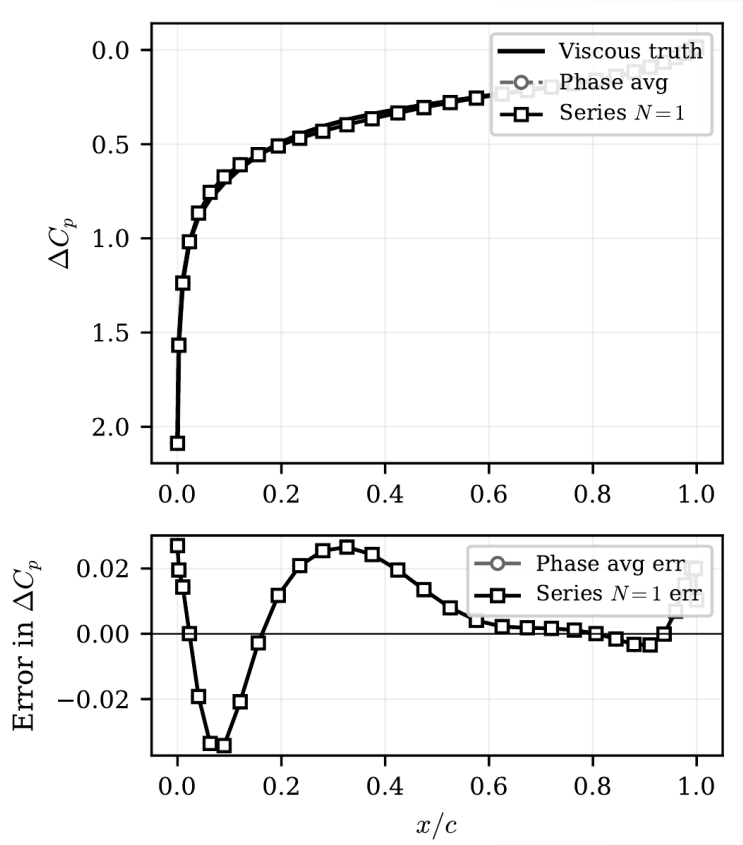
After our correction:

$\approx 10^{-16}$ to 10^{-13}

Algebra check passed: error at rounding level

p. 3, Table 2 footnote; p. 4, Sec. 5.1 and Figs. 1–2

Viscous test: one amplitude alone gets worse as it grows



Loading along the chord at 2°; error below

pp. 3–4, Table 2 and Secs. 4.2–4.4; p. 5, Fig. 3 and Table 3

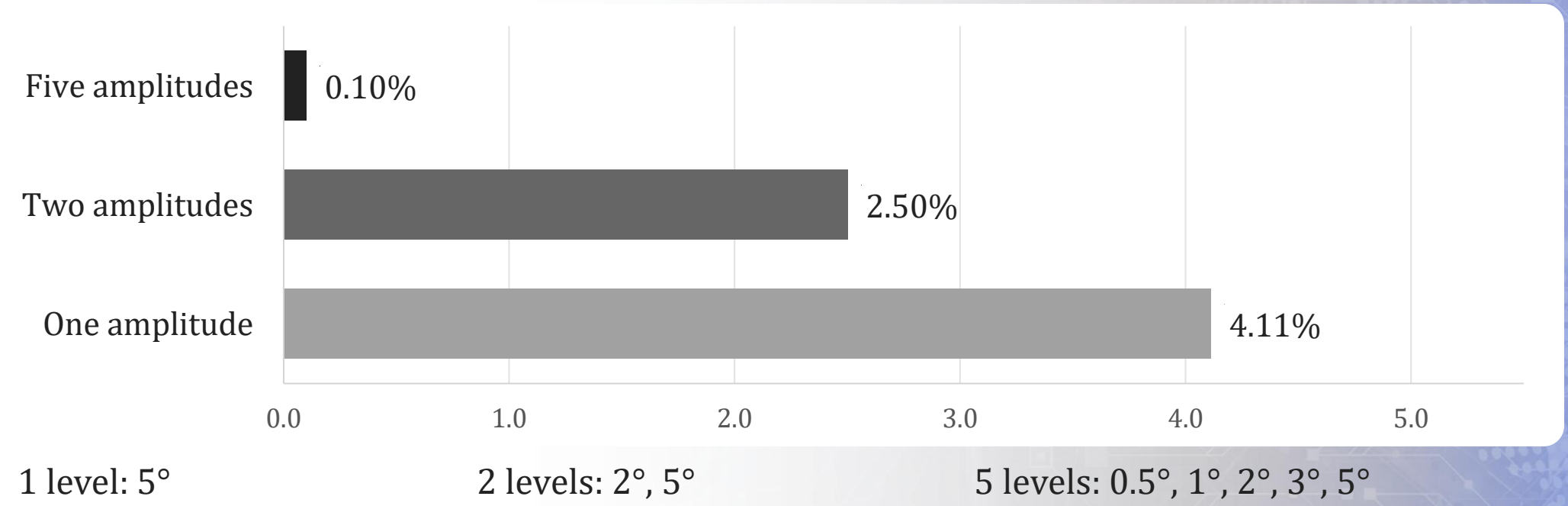
NACA 0012 (symmetric airfoil)

$$\alpha_0 = 4^\circ, \quad \mathrm{Re} = 10^6$$

Amplitude (°)	Relative L^2 error
0.5	0.44%
1.0	1.30%
2.0	2.58%
3.0	3.33%
5.0	**4.11%**

Error of one amplitude alone. Average and fit agree.

Combining five amplitudes cuts the error to 0.10%



1 level: 5°

2 levels: 2°, 5°

5 levels: 0.5°, 1°, 2°, 3°, 5°

Five amplitude levels: 40-fold lower error than one level at 5°

Error in % (lower is better); slow-motion test with the NeuralFoil model

p. 5, Table 3 and Sec. 5.2

Summary: two steps recover the static loading at one angle

Two steps

1 Speed-gradient operator: fit each record, keep the constant

2 Combine several amplitudes to cancel the mean bias

0.10% relative error

with five amplitudes in the viscous-model test (slow motion)

pp. 3–6, Secs. 3.3, 5, and 6

Next steps: fast-motion viscous data and several angles

Validation

Test on viscous data with fast motion (wake memory)

Convergence and resolution

Viscous-test error grows like ε, not ε^2: check with a finer grid

Scope

Extend from one mean angle to several

p. 5, Sec. 5.2; p. 6, Sec. 6.0.1

Thank you

Questions & discussion

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